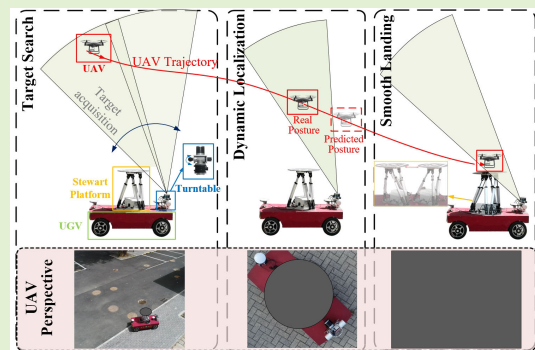


LCDL: Toward Dynamic Localization for Autonomous Landing of Unmanned Aerial Vehicle Based on LiDAR–Camera Fusion

Yongkang Xu^{ID}, Graduate Student Member, IEEE, Zhihua Chen^{ID}, Chencheng Deng, Shoukun Wang^{ID}, and Junzheng Wang^{ID}

Abstract—The detection, localization, and tracking of unmanned aerial vehicles (UAVs) are pivotal for ensuring reliable decision-making and intelligent control in scenarios involving heterogeneous agent cooperation. A dynamic localization framework with asynchronous LiDAR–camera fusion is considered in this article, which is used to provide absolute attitude and position observations of UAV and to achieve robust localization in outdoor environments. First, a fast search architecture based on the depth cluster is presented to transform point clouds into distance images and establish distance image target extraction based on any two neighboring points. Besides, a neural network framework is introduced for the recognition of UAVs, where the feature maps are fed into a region suggestion network to obtain optimal suggestions for object classification and bounding box regression. Furthermore, we designed a dual servo turntable integrated with multisensors to dynamically track the coordinates of the UAVs, ensuring that the vehicle remains centered within the detection area at all times. Finally, the heterogeneous agent is employed to evaluate the localization performance of UAVs in real-world situations. This indicates that asynchronous LiDAR–camera fusion can run fully on embedded devices and productively in heterogeneous agent systems.

Index Terms—Asynchronous sensor, dynamic localization, information fusion, unmanned aerial vehicles (UAVs).



I. INTRODUCTION

LANDING remains widely recognized as one of the most challenging tasks for unmanned aerial vehicles (UAVs) [1], [2], [3]. This complexity arises from the

Manuscript received 12 March 2024; accepted 10 June 2024. Date of publication 16 July 2024; date of current version 15 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62173038, in part by the Fundamental Research Funds for the Central Universities of China under Grant 2024CX06030, in part by the Foundation of Key Laboratory of Jiangxi Province for Image Processing and Pattern Recognition under Grant ET202204304, in part by Jiangxi Natural Science Foundation under Grant 20232BAB212028, in part by the Foundation of Science and Technology of Jiangxi Provincial Education Department under Grant GJJ2201128, and in part by the Doctor's Foundation of Nanchang Hangkong University under Grant EA202204257. The associate editor coordinating the review of this article and approving it for publication was Prof. Bin Gao. (Corresponding author: Shoukun Wang.)

Yongkang Xu, Chencheng Deng, Shoukun Wang, and Junzheng Wang are with the National Key Laboratory of Autonomous Intelligent Unmanned Systems, School of Automation, Beijing Institute of Technology, Beijing 100081, China (e-mail: bitwsk@bit.edu.cn).

Zhihua Chen is with the Key Laboratory of Jiangxi Province for Image Processing and Pattern Recognition and MOE Key Laboratory of Nondestructive Testing Technology, Nanchang Hangkong University, Nanchang 330063, China (e-mail: zhihua.chen@nchu.edu.cn).

Digital Object Identifier 10.1109/JSEN.2024.3424218

requirement for UAVs to achieve real-time and precise location information up to the point of landing [4], [5], [6]. The autonomous operation of these vehicles in outdoor environments primarily relies on the global navigation satellite system (GNSS) [7]. This approach offers the advantage of estimating absolute coordinates based on GNSS receiver equipment. However, GNSS signals face susceptibility to blockage by buildings, vulnerability to disruption, and potential exposure to interference and deception.

Crucial research directions, which focus on small targets in the nearby airspace, encompass precise recognition and swift localization within the UAV ground autonomous recovery and landing field. The airborne sensor localization method is studied for UAV active localization, and a method to determine the position of objects in the world by using the fixed-wing UAV airborne general camera was proposed by Redding et al. [8]. AprilTag recognition is employed to complete the localization of the UAV, and visual guidance is used to complete the autonomous landing [9].

To enhance the lightweight design of UAVs, researchers are progressively investigating ground multisensor configurations for the passive localization of UAVs. A scheme is proposed

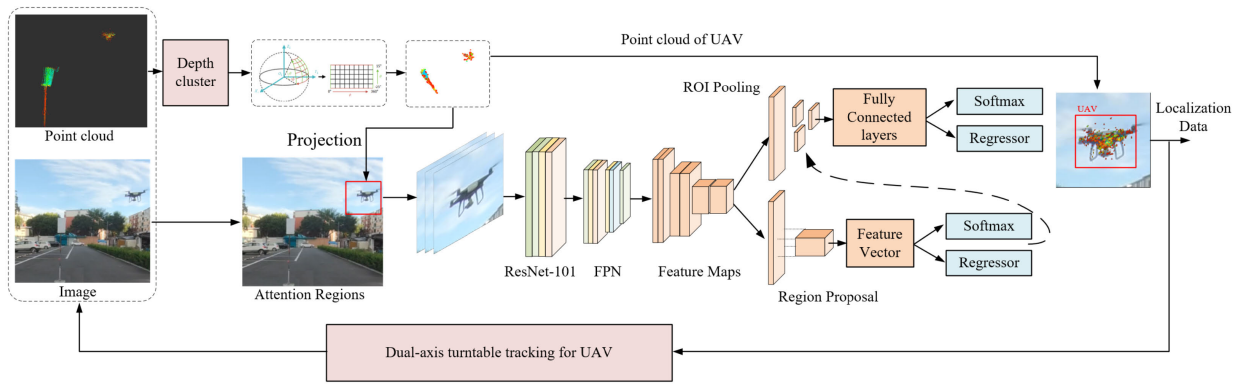


Fig. 1. Localization architecture of the proposed asynchronous LiDAR-camera fusion method with real-time attention guidance.

to erect a camera pan-tilt-zoom unit on the ground for the localization of fixed-wing UAVs, and the AdaBoost method is used to detect and track UAVs [10]. High-power LED lights with active markings of different colors are installed at the wing end of the quadrotor UAV, and an onboard stereo camera setup composed of two webcams is used to provide accurate location information. The error within 1 m of the UAV from the localization system can be controlled to within 15 cm [11]. The data fusion of two infrared active mode cameras and ultrasonic sensors is used to accurately estimate the position of the UAV in [12]. The unmanned ground vehicle (UGV) with four UWB transceivers and the UAV equipped with two transceivers form a localization system in [13]. The process is initiated by the UAV transceiver, which sends signals to each UGV transceiver in sequence. The average error of the target within 30 m in the X - and Y -directions is 1 m, while the average error in the Z -direction is 20 cm.

The LiDAR-camera fusion, which is widely adopted, can acquire precise spatial coordinates of the target, thereby possessing significant application value for localization. In RGB images, 2-D object region proposals are generated by convolutional neural networks, which are subsequently extended into a 3-D visual cone, as discussed in [15]. The 3-D bounding box is predicted by Frustum PointNet, leveraging the points within the visual cone. In [16], the generated three-channel point cloud tensor is combined with RGB images to form a six-channel tensor. A Gaussian YOLO network structure is employed for active vision based on object detection. In [17], 3-D bounding boxes were generated by fusing the LiDAR point cloud with camera images. The point cloud, which was enhanced solely by image feature maps at full resolution, was matched with multimodal data using window convolution. Subsequently, new features were extracted from the enhanced point cloud for the purpose of generating bounding boxes. Algorithms based on deep learning include the voxel network, which utilizes a single-stage architecture, the point cloud columnar feature network, and the point-based graph neural network (Point-GNN [18]). In addition, two-stage detection methods are also employed—the point cloud voxel region-based convolutional neural network (PV-RCNN [19]), voxel region-based convolutional neural network (Voxel R-CNN [20]), and other algorithms. At the

same time, point cloud target detection based on deep learning requires high-computing resources and has poor real-time performance on platforms with limited computing power.

To address the identified research gap and propel advancements in UAV localization, this study introduces an asynchronous LiDAR-camera fusion localization framework. A rapid search process incorporating deep clustering has been devised to enhance the learning efficiency and performance of Faster R-CNN. In each localization step, ego-motion compensation aligns the turntable actions based on their respective utilities. The main contributions of this article are summarized as follows.

- 1) An asynchronous LiDAR-camera fusion technique was designed, which ensures that position and attitude observations are continuously available even when GNSS signals may fail.
- 2) This study introduces the attention mechanism to enhance localization optimization. Subsequently, the system incorporates this optimized localization technology featuring the attention mechanism, resulting in improved accuracy and a reduction in time costs.
- 3) Recognizing the inherent limitations of sensor detection ranges, we designed a two-axis carousel specifically for focused and precise tracking of UAVs by integrating a model predictive control (MPC) approach.

The remaining structure of this article is as follows. Section II introduces the multisensor fusion method in this experiment. Section III designed an actual testing scenario and analyzed the localization results of the asynchronous LiDAR-camera fusion. Finally, Section IV summarizes the results of this article and provides prospects for future work.

II. METHODOLOGY

The real-time localization flowchart for UAV via asynchronous LiDAR-camera fusion proposed in this article is shown in Fig. 1. At the outset, LiDAR point cloud and image data are gathered and synchronized. Subsequently, a moving UAV is swiftly identified through depth clustering of the point cloud. In addition, the UAV's point cloud is projected onto the camera's image plane, delineating an attention area for image recognition and UAV target detection. Third, UAV localization is conducted by amalgamating the LiDAR real point cloud

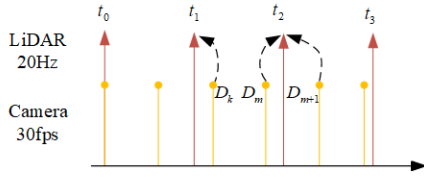


Fig. 2. Different sampling time of LiDAR and camera.

with position information in the image coordinate system. Finally, to ensure that sensors mounted on the servo turntable consistently remain within the optimal detection range, a servo turntable tracking control method based on MPC is devised. This method achieves focused sensor tracking of aerial targets by controlling the azimuth and pitch servo motors of the turntable.

A. Image and Point Cloud Data Synchronization

1) *Timestamp Synchronization*: To guarantee that each sensor operates under a consistent time reference, it is imperative to explore effective methods for time synchronization, taking into account potential disturbances, such as inconsistent clock references for each sensor and varying data frequencies within the asynchronous multisensor system.

In this article, interpolation or estimation methods are used to align the above perceptual data with the time base of the central computer, as shown in Fig. 2. For example, the point cloud data and image sensor data at time t_1 are fused by direct interpolation; that is, the image D_k with the closest sampling exposure time to t_1 is aligned and fused. At time t_2 , the point cloud data and the image sensor data are fused using a linear weighting method—the t_m time exposure image data D_m and the $t_m + 1$ time exposure image data D_{m+1} . The exposure time of these two frames of image data is the closest time from time t_2 . That is, D_m data are used as image data at time t_2 .

2) *Space Synchronization*: Taking the joint calibration of LiDAR and camera as spatial synchronization, with $o-x-y-z$ as the LiDAR coordinate system, $O-X-Y-Z$ as the camera coordinate system, and $o-u-v$ as the pixel coordinate system, the coordinate conversion matrix T_L^C from the LiDAR coordinate system to the camera coordinate system can be obtained through the joint calibration of the LiDAR and the camera, so the coordinate point $P_L(x, y, z)$ under each LiDAR coordinate can be converted into point $P_C(x, y, z)$ in the camera coordinate system. The LiDAR–camera calibration calculates the extrinsic matrix T_L^C to minimize the reprojection error between image pixels $G_{P_{uv}}$ and the matched projected pixels $G_{\hat{P}_{uv}}$ of the LiDAR point $P_L(x, y, z)$, which is formulated as follows:

$$\min \sum_i^m \|G_{P_{uv}} - G_{\hat{P}_{uv}}\|^2. \quad (1)$$

The target-based method, which commonly employs 3D–3D point correspondences [22], is utilized for extrinsic calibration to calibrate the extrinsic transformation between the LiDAR and the camera. The rectangular cardboard and ArUco mark are found by the 3D–3D correspondence between the

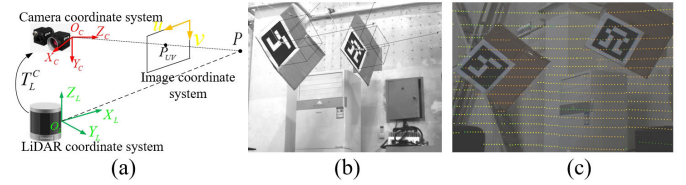


Fig. 3. Space synchronization results. (a) Coordinate system space transformation relationship. (b) ArUco markers. (c) Space synchronization results.

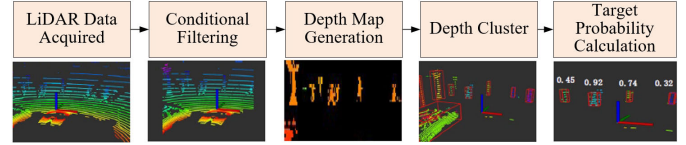


Fig. 4. Architecture of the proposed fast search method with depth cluster. Introducing an range image projection into both real-time depth cluster.

point cloud and the camera image, as shown by the coordinate system transformation relationship in Fig. 3(a) and the synchronization results in Fig. 3(b) and (c). The rotation and translation transformation between the board's frame-of-reference and the camera's frame-of-reference will be found by the ArUco marker, which is used to transform the 3-D corner from the plate frame to the camera frame.

B. Recognition for UAV by LiDAR

To address the challenge of low real-time performance in LiDAR and localization calculations, particularly when handling massive high-resolution point cloud data, we propose a target segmentation method based on depth clustering. This method enables a swift search of the target area. The operational steps of the rapid target search method, grounded in deep clustering, are illustrated in Fig. 4. The original 3-D point cloud data are conditionally filtered and then projected onto a front view to be represented as a depth map, a mapping relationship between the depth map and the point cloud data is established, a fast depth clustering algorithm based on the depth map and self-adaptive adjustment of clustering radius is designed to cluster the point cloud to obtain a target candidate region, and the target candidate region with the highest probability is obtained through a multidimensional feature operator.

Aiming at the problem of low efficiency of traditional clustering algorithm, a clustering segmentation algorithm based on calculating the depth angle of 3-D point cloud is proposed. Complete the segmentation of different objects to obtain the clustering target. The principle is shown in Fig. 5.

After completing the point cloud clustering, the prerequisite for UAV localization is to detect the UAV. We combine the features of the UAV with the LiDAR data and design an evaluation function to assess the objects clustered from the point cloud for UAV recognition, which is formulated as follows:

$$f_d = d_I \cdot \frac{I_T - I_{th}}{I_{th}} + d_V \cdot \frac{V_T - V_{th}}{V_{th}} + d_P \cdot \frac{P_T - P_{th}}{P_{th}} \quad (2)$$

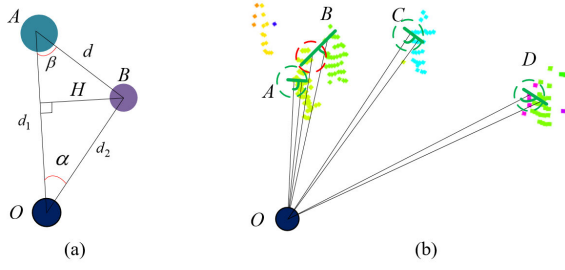


Fig. 5. Depth cluster method of the proposed range image projection with point cloud. (a) It illustrates the shape formed by any two adjacent points in the range image and the LiDAR coordinate origin. By calculating the angle β to be determined whether two points belonged to the same object, where A and B are two points collected by the LiDAR, d denotes the distance between A and B , and d_1 denotes the longer distance between OA and OB . (b) Depth cluster segmentation of pedestrians captured by LiDAR from the Nuscenes dataset is demonstrated.

Algorithm 1 Label Range Image

```

INPUT  $R \leftarrow$  range image
OUTPUT  $L \leftarrow$  label image
for  $pt = \text{iterator}(R)$  do
   $\text{queue.push}(pt)$ 
  while  $\text{queue is not empty}$  do
     $\text{queue.pop}$ 
    for  $pt_n = \text{Neighbor}(pt)$  and  $pt_n$  is Unknown do
       $d_1 \leftarrow \max(pt.\text{depth}, pt_n.\text{depth})$ 
       $d_2 \leftarrow \min(pt.\text{depth}, pt_n.\text{depth})$ 
       $\beta = \arccos \frac{d^2 + d_1^2 - d_2^2}{2dd_1}$ 
      if  $\beta > \beta_{th}$  or  $d < d_{th}$  then
         $\text{queue.push}(pt_n)$ 
         $L(pt_n) \leftarrow \text{Label}$ 
      end
    end
  end
   $\text{Label} \leftarrow \text{Label} + 1$ 
end

```

where d_I , d_V , and d_P are weighting parameters, where I represents the average reflectivity of the object, V represents the volume of the bounding box of the object, and P represents the position of the object. Subscripts with th indicate thresholds, and subscripts with T indicate the parameters of the clustered objects measured at the current time.

C. Recognition for UAV by Camera

UAV recognition algorithm based on deep learning is the mainstream method at present. It can be mainly divided into two categories, one is a single-stage algorithm, which directly obtains the target position and category through an end-to-end network, such as the YOLO series [23], [24], [25], [26]. The other is a two-stage algorithm, which first generates a series of the target candidate region that is classified, and then, the candidate region is classified, such as the R-CNN [27], [28], [29]. This article introduces a UAV recognition model founded on the Faster R-CNN framework. The architecture comprises three fundamental layers: a pretrained deep convolutional neural network (DCNN), a regional proposal network (RPN),

and a fully connected layer (FC). The Faster R-CNN model exhibits limitations in accurately detecting small targets due to the UAV's distance from the camera. To address this, the VGG-16 convolutional network within the Faster R-CNN network is substituted with the simpler and deeper ResNet101. This substitution effectively mitigates issues associated with gradient dispersion and gradient explosion in the VGG network.

Compared with the traditional YOLOv5 algorithm, which excels in recognition, our combined algorithm demonstrates superior performance in identifying distant objects. Where in our combined algorithm, in the case of a simple background, the frame difference method can detect moving objects even if the objects are very small. Faster R-CNN uses anchors to generate proposals at each sliding window location, simultaneously. The k suggestion is a parameterized suggestion for the k reference frame anchor. Each anchor point is centered around the sliding window and associated with scale and ratio. Faster R-CNN uses three scales and three aspect ratios, with a total of nine anchors per sliding position. An important feature of this method is translation invariance. Therefore, it is suitable for situations where the object size changes.

The soft rims algorithm can optimize the confidence of the anchor frame. Each time the anchor frame with the highest confidence is selected, a Gaussian function is introduced to suppress the confidence of the surrounding anchor frame and reduce the missed detection. The Gaussian function formula is

$$s_i = s_i e^{-\frac{\text{IOU}(M, b_i)^2}{\sigma}} \quad \forall b_i \in D \quad (3)$$

where M is the box with the highest score at present and is the box to be processed, b_i is the i th detection frame, s_i is the score of the i th detection frame, and σ is 0.5 for the super parameter. After the soft-NMS algorithm is used to optimize the confidence of the anchor frame, the accuracy of the model prediction can be improved, and the occurrence of target loss can be reduced when multiple small targets overlap.

In this article, average precision (AP) and average recall (AR) are mainly used to measure the detection performance of the network for UAV targets. First, calculate the precision and recall. The calculation formula is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (5)$$

where TP is the positive class predicted as the number of positive classes, that is, the number of detection frames with $\text{IOU} \geq 0.5$; FP is the number of negative classes predicted as positive classes, that is, the number of detection frames with $\text{IOU} \leq 0.5$, or the number of redundant detection frames when the same real frame is detected; and FN is the number of positives predicted as negative classes.

D. Localization for UAV by Asynchronous LiDAR–Camera Fusion

After the image UAV recognition is completed, the area of the UAV in the image is obtained non-Maxows

$$b_u = (u_u, v_u, w_u, h_u) \quad (6)$$

where u_u and v_u are the pixel coordinates of the target in the top-left corner of the frame in the image, and w_u and h_u are the width and height of the border of the target in the image.

The external parameter T_L^C is obtained by the camera's internal parameter K and calibrated by the camera and the LiDAR. The n point cloud P_i^L is projected into the image coordinate system through PP to obtain the projected point $P_i^{UV} = (u_i, v_i)$. Then, the point that does not meet the conditions of formula (6) is removed; that is, the point clouds outside the target frame are removed, and only the point clouds inside the target frame are retained, to reduce the number of point clouds and reduce the demand for computing resources for subsequent processing. So far, the fusion of point cloud and image data is completed as follows:

$$\begin{cases} u_k < u_i < (u_k + w_k) \\ v_k < v_i < (v_k + h_k) \end{cases}, \quad i = 1, 2, \dots, n, \quad k = 1, 2, \dots, m \quad (7)$$

where u_i and v_i represent the coordinates of the pixel point after the point cloud is projected to the image coordinate system. The figure shows the result after data fusion. Image target detection can only provide the approximate area of the target. There is a certain background in the target frame, and the target is easily affected by occlusion. There will be some point clouds in the target frame that do not belong to the target, which will affect the accuracy of target localization. These point clouds are discriminated, and the real point cloud of the target is extracted.

E. Tracking for UAV by Dual-Axis Turntable

The ego-motion compensation is a coordinated conversion process. Correspondingly, a set of data A is collected at time t_A when a UAV is located at $(x, y, z, \theta, \psi, \phi)$, the UAV flight to $(x + \Delta x, y + \Delta y, z + \Delta z, \theta + \Delta \theta, \psi + \Delta \psi, \phi + \Delta \phi)$ at time $t_A + \Delta t$, and the dataset is marked as A' . The objective of the ego-motion compensation step is to calculate this transformation T_m . The transformation matrix can be estimated directly, which infers the transformation T_m from the corresponding significant features in dataset A to the dataset A' .

The Kanade–Lucas–Tomasi (KLT) corner detector feature tracking algorithm was developed to guarantee computational efficiency. It selects a set of features F^{t-1} from continuous images (time $t - 1$) and constructs the same features on the corresponding feature set F^t (current time t) by tracking. The model parameters of ego-motion compensation are estimated by minimizing the cost, and the cost function is defined as follows:

$$C = \frac{1}{2} \sum_{i=1}^m (F_i^t - T_{t-1}^t F_i^{t-1})^2 \quad (8)$$

where T_{t-1}^t is transformation from F^{t-1} to feature set F^t and m is the total number of sets of features.

The dual-degree-of-freedom turntable achieves servo control of the UAV rotation angle by governing the servo motor installed on the rotational axis, constituting essentially a motor position servo control process. The schematic illustrating UAV

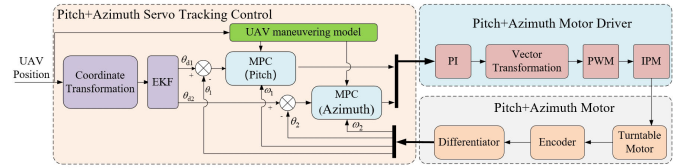


Fig. 6. UAV tracking by ego-motion compensation.

tracking through ego-motion compensation is depicted in Fig. 6. A tracking differentiator is added to the motor servo control module to improve the dynamic characteristics of the turntable and obtain the target angular velocity ω_t , which is superimposed on the motor speed target command as a speed feedforward component. MPC is adopted as a controller to improve the tracking performance of the system through model prediction, rolling optimization, and feedback correction.

Since the mathematical expression of the linear motor is second order, using two-step prediction, the objective function can be expressed as follows:

$$J = \sum_{i=1}^n [\xi(k+i) - \omega(k+i)]^2 + \sum_{i=1}^m \lambda u^2(k+i-1) \quad (9)$$

where ω is the reference input signal, n is the maximum prediction length, m is the control length, and λ is the weighting coefficient. The objective function, denoted as J , aims to calculate differential equations $u(k)$ and $u(k+1)$, making it equal to 0. The respective figure illustrates this process. Subsequently, the value of the controller output in the next step can be determined. The first element of the sequence is used as the actual control input to act on the system, and after entering the next control cycle, the above process is repeated to complete the control of the servo turntable track.

III. EXPERIMENTAL VALIDATION

A. Experiment Overview

To assess the effectiveness and feasibility of the proposed localization method employing asynchronous LiDAR–camera fusion, a series of experiments were conducted, specifically designed for heterogeneous UAV–UGV cooperative tracking and landing scenarios, incorporating asynchronous LiDAR–camera fusion. Specifically, the experimental validation platform and corresponding scenarios are illustrated in Fig. 7(a) and (b). This integration leverages the precision, speed, and stability offered by a multisensor device, an all-terrain UGV, and a Stewart parallel platform.

The multisensor device includes a LiDAR, camera, infrared camera, radar, IMU, and dual-axis turntable. The LiDAR, with the model RS-LiDAR-32, is a long-range sensor. These parameters encompass the sampling rate, LiDAR wavelength, field angle, detection precision, and target discrimination range. The LiDAR data consist of target position information with a timestamp but are initially disordered. For target recognition, the BFLY-PGE-13E4M/C-CS camera is employed, as it is a prerequisite for the asynchronous LiDAR–camera fusion process to identify targets. To facilitate night-time target recognition, we have incorporated the infrared camera model DS-2TM03-LF/AN. This selection aims to bolster the



Fig. 7. Experimental verification platform. (a) Experimental setup employed in this study consists of an integrated two-degree-of-freedom servo turntable equipped with LiDAR, a camera, and an infrared camera. (b) Localization experiment for UAV was provided to test the localization performance of the localization architecture obtained by different methods. UAV is used in this experiment, and the wheelbase of this model is 350×350 mm.

capability of identifying UAVs in conditions of low light or during the night.

The information processing computer utilizes the Nuvo-7160 series, a high-performance embedded computer. It is capable of supporting a GPU with a power consumption of up to 120 W, delivering computing power in the range of 4–6 TFLOPS. Moreover, it is equipped with an Intel ninth Gen 16-core CPU and features an NVIDIA GTX 1660Ti graphics card.

The experimental scenario designed a challenging localization task, which involves the drone continuously flying different trajectories. To guarantee successful completion of design tasks in all scenarios, the UGV and UAV were required to start from a fixed position and ultimately achieve high-precision localization of UAV through asynchronous LiDAR–camera fusion. In the multiple sets of experiments designed, UAV was requested to provide line and in an S-curve flight trajectories for the designed asynchronous LiDAR–camera fusion and other methods—that is, YOLOv3, YOLOv5, and Euclidean cluster by LiDAR.

B. Accuracy Performance on the Localization

During the experiments, the positions of the UAVs were recorded and statistically analyzed for each method to assess the localization performance. The results, depicted in Fig. 8 from the UAV localization experiments, demonstrate the enhanced performance of the proposed asynchronous LiDAR–camera fusion method compared with other state-of-the-art localization algorithms, such as YOLOv3, YOLOv5, and Euclidean clustering, leveraging LiDAR. As illustrated in Fig. 9, the outcomes indicate that the turntable tracking method surpasses other approaches in terms of field of view angle and positioning efficiency. Building upon the previous research, scores are reported as evaluation metrics, which are displayed in Table I.

The UAV was designed to navigate in both straight-line paths and S-curves while in the air, with data recorded from each sensor. Fig. 10 displays the dynamic localization curves of the asynchronous LiDAR–camera fusion system for the actual position of the UAV, i.e., the localization data in the upper X-, Y-, and Z-directions during straight-line motion. The red, blue, and green curves represent the camera, LiDAR,

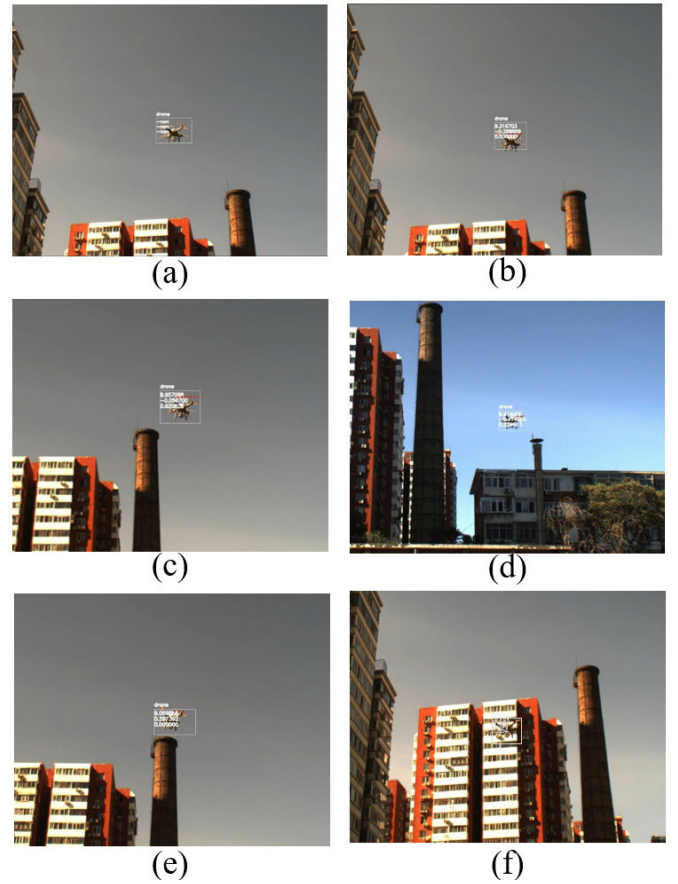


Fig. 8. Localization experimental results for quadrotor UAV. (a)–(e) UAV localization against a clear background. (f) UAV localization against a backdrop of buildings.



Fig. 9. Localization experimental results for UAV tracking by ego-motion compensation. (a) Initial position of the turntable. (b) Servo turntable follows the tracks of the UAV.

TABLE I
INSPECTION RESULTS ON THE UAV DATASET

Detector	mAP	$AP(f_{ray})$
PointNet [30]	65.75	53.64
AdaFusion [31]	77.74	73.58
PIC-Net [32]	84.36	76.55
DeepFusion [33]	81.24	75.17
LCDL(ours)	89.75	80.49

and asynchronous LiDAR–camera fusion localization errors for UAV, respectively.

Fig. 11 shows the experimental results of the UAV’s actual localization. Fig. 11(a)–(d) shows the statistical analysis of the experimental data. The results show that the asynchronous LiDAR–camera fusion method was advantageous

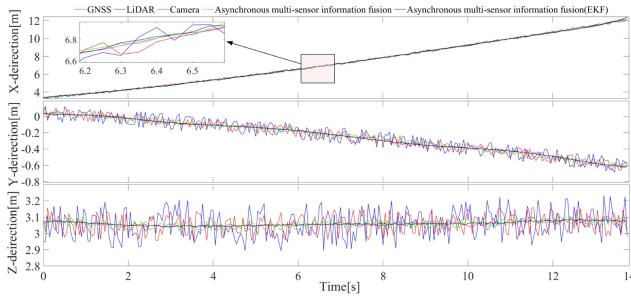


Fig. 10. Localization trajectory of the UAV is analyzed under straight flight conditions.

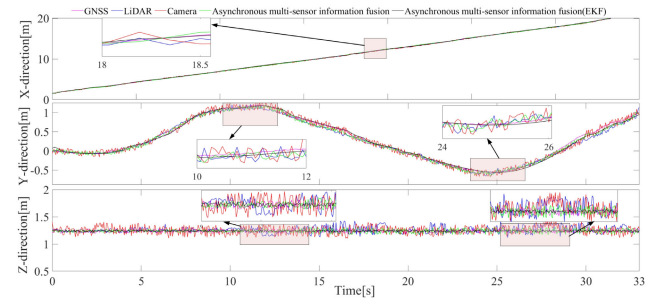


Fig. 12. Localization trajectory of the UAV under curved flight conditions.

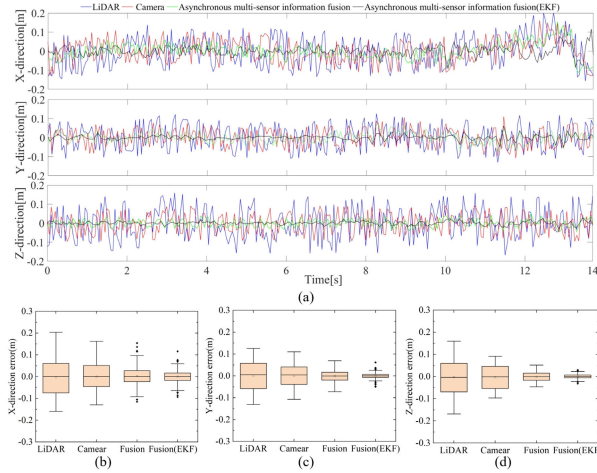


Fig. 11. (a) Comparison of the localization errors of the UAV in the X-, Y-, and Z-directions of the linear trajectory. (b) Box plots of the X-direction localization errors for UAV under the provided by LiDAR. (c) Box plots of the X-direction localization errors for UAV under the provided by camera. (d) Box plots of the X-direction localization errors for UAV under the provided by asynchronous LiDAR-camera fusion.

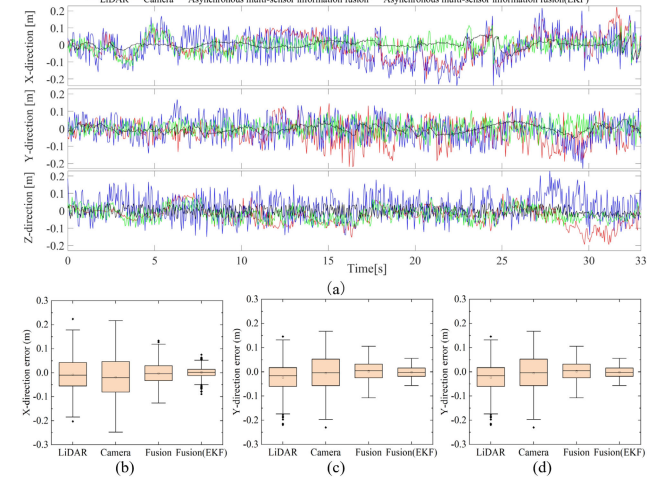


Fig. 13. (a) Comparison of the localization errors of the UAV in the X-, Y-, and Z-directions of the S-shaped trajectory. (b) Box plots of the X-direction localization errors for UAV under the provided by LiDAR. (c) Box plots of the X-direction localization errors for UAV under the provided by camera. (d) Box plots of the X-direction localization errors for UAV under the provided by asynchronous LiDAR-camera fusion.

over all other basic methods with respect to localization accuracy. Fig. 11(a) indicates that the localization accuracy obtained with the asynchronous LiDAR-camera fusion during the localization experimental was the highest (X -direction: $Mr = 0.0004$ and $SDr = 0.0272$, Y -direction: $Mr = 0.0001$ and $SDr = 0.0122$, and Z -direction: $Mr = 0.0003$ and $SDr = 0.0095$, where Mr and SDr are the average and standard deviation of the localization error, respectively), and then, there is the acquisition with the YOLOv5 method (X -direction: $Mr = 0.0007$ and $SDr = 0.0623$, Y -direction: $Mr = 0.0007$ and $SDr = 0.0759$, and Z -direction: $Mr = -0.0025$ and $SDr = 0.0543$) and then the Euclidean cluster (X -direction: $Mr = -0.0028$ and $SDr = 0.0806$, Y -direction: $Mr = 0.0018$ and $SDr = 0.065$, and Z -direction: $Mr = -0.002$ and $SDr = 0.0828$).

The experimental results of the actual positioning of the UAV flying according to the S-curve are shown in Fig. 12, and the dual-axis turntable tracks the UAV in real time in the entire localization experimental process. The S-curve localization experimental statistical results shown in Fig. 13(a)–(d) illustrate that the localization accuracy with the asynchronous LiDAR-camera fusion of UAV S-curve flight trajectory was the highest (X -direction: $Mr = 0.0018$ and $SDr = 0.0231$, Y -direction: $Mr = -0.0013$ and $SDr = 0.0227$, and Z -direction: $Mr = -0.0026$ and $SDr = 0.0249$),



Fig. 14. Autonomous landing experiments for UAV.

the YOLOv5 method (X -direction: $Mr = 0.0186$ and $SDr = 0.0844$, Y -direction: $Mr = -0.0050$ and $SDr = 0.0698$, and Z -direction: $Mr = 0.0175$ and $SDr = 0.0730$), and then the Euclidean cluster (X -direction: $Mr = -0.0078$ and $SDr = 0.069$, Y -direction: $Mr = -0.023$ and $SDr = 0.0642$, and Z -direction: $Mr = -0.0030$ and $SDr = 0.055$).

As shown in Fig. 14, the method has achieved dynamic localization of the UAV and enabled real-time tracking of the UGV based on the relative position of the UAV. Finally, the UAV can safely landing on the UGV.

IV. CONCLUSION

This article introduces a localization architecture for accurate UAV positioning on a UGV-UAV recovery landing platform in complex environments, utilizing asynchronous LiDAR-camera fusion. The validated asynchronous

LiDAR–camera fusion-based localization was subjected to localization experiments and compared with other state-of-the-art localization approaches. The experimental results indicate that the proposed asynchronous LiDAR–camera fusion surpasses the existing methods in localization performance. This method effectively enhances the search and localization performance for UAVs in practical environments. Through experimental results on UAV tracking, the performance of the proposed fusion estimator has been showcased, demonstrating a broad detection range and producing reliable fused estimates. The standard deviations of UAV localization achieved through asynchronous LiDAR–camera fusion are 10.5 mm, representing a significant improvement compared with the results obtained using YOLOv3, YOLOv5, and the Euclidean cluster method with LiDAR. The elevated framework, methods, and algorithms formulated in this study possess significant potential for extension to a diverse array of applications involving UGV–UAV landing interactions.

REFERENCES

- [1] Q. Tang et al., “Physical coupling fusion sensing of MFL-EMAT for synchronous surface and internal defects inspection,” *IEEE Sensors J.*, vol. 23, no. 14, pp. 16068–16079, Jul. 2023.
- [2] H. Wen, B. Gao, D. Yang, Y. Zhang, L. Huang, and W. L. Woo, “Wearable integrated online fusion learning filter for heart PPG sensing tracking,” *IEEE Sensors J.*, vol. 23, no. 13, pp. 14938–14949, Jul. 2023.
- [3] J. Cao, N. Wang, S. Qian, and W. Guan, “When to invoke a prediction service for Bus. Process monitoring?” *IEEE Trans. Services Comput.*, vol. 16, no. 4, pp. 1–14, Feb. 2023.
- [4] G. Zhu, Y. Yang, R. Sun, E. Q. Wu, and R. Law, “A similarity measurement method of normal cloud models for the operational status perception and computing of urban rail transit,” *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 1, pp. 746–755, Feb. 2024.
- [5] G. Zhu, Y. Gong, J. Ding, E. Q. Wu, and R. Law, “Time granularity setting principle for short-term passenger flow prediction in urban rail transit,” *IEEE Trans. Computat. Social Syst.*, early access, May 9, 2024, doi: [10.1109/TCSS.2024.3385850](https://doi.org/10.1109/TCSS.2024.3385850).
- [6] G. Zhu, J. Ding, Y. Wei, Y. Yi, S. S. Xu, and E. Q. Wu, “Two-stage OD flow prediction for emergency in urban rail transit,” *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 920–928, Jan. 2024.
- [7] N. Gyagenda, J. V. Hatilima, H. Roth, and V. Zhmud, “A review of GNSS-independent UAV navigation techniques,” *Robot. Auto. Syst.*, vol. 152, Jun. 2022, Art. no. 104069.
- [8] J. D. Redding, T. W. McLain, R. W. Beard, and C. N. Taylor, “Vision-based target localization from a fixed-wing miniature air vehicle,” in *Proc. Amer. Control Conf.*, 2006, pp. 2862–2867.
- [9] C. S. Sharp, O. Shakeria, and S. S. Sastry, “A vision system for landing an unmanned aerial vehicle,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2001, pp. 1720–1727.
- [10] W. Kong et al., “A ground-based optical system for autonomous landing of a fixed wing UAV,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Sep. 2014, pp. 4797–4804.
- [11] M. Achtelik, T. Zhang, K. Kuhnlenz, and M. Buss, “Visual tracking and control of a quadcopter using a stereo camera system and inertial sensors,” in *Proc. Int. Conf. Mechatronics Autom.*, Aug. 2009, pp. 2863–2869.
- [12] I. Kalinov, E. Safronov, R. Agishev, M. Kurenkov, and D. Tsetserukou, “High-precision UAV localization system for landing on a mobile collaborative robot based on an IR marker pattern recognition,” in *Proc. IEEE 89th Veh. Technol. Conf. (VTC-Spring)*, Apr. 2019, pp. 1–6.
- [13] Y. Xianjia, L. Qingqing, J. P. Queralt, J. Heikkinen, and T. Westerlund, “Cooperative UWB-based localization for outdoors positioning and navigation of UAVs aided by ground robots,” in *Proc. IEEE Int. Conf. Auton. Syst. (ICAS)*, Aug. 2021, pp. 1–5.
- [14] Y. Zhou and O. Tuzel, “VoxelNet: End-to-end learning for point cloud based 3D object detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4490–4499.
- [15] C. R. Qi, W. Liu, C. Wu, H. Su, and L. J. Guibas, “Frustum PointNets for 3D object detection from RGB-D data,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 918–927.
- [16] C. Shi, G. Lai, Y. Yu, M. Bellone, and V. Lippiello, “Real-time multi-modal active vision for object detection on UAVs equipped with limited field of view LiDAR and camera,” *IEEE Robot. Autom. Lett.*, vol. 8, no. 10, pp. 6571–6578, Oct. 2023.
- [17] C. Nie, Z. Ju, Z. Sun, and H. Zhang, “3D object detection and tracking based on LiDAR-camera fusion and IMM-UKF algorithm towards highway driving,” *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 7, no. 4, pp. 1242–1252, Aug. 2023.
- [18] W. Shi and R. Rajkumar, “Point-GNN: Graph neural network for 3D object detection in a point cloud,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1711–1719.
- [19] S. Shi et al., “PV-RCNN: Point-voxel feature set abstraction for 3D object detection,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10529–10538.
- [20] J. Deng et al., “Voxel R-CNN: Towards high performance voxel-based 3D object detection,” in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2021, vol. 35, no. 2, pp. 1201–1209.
- [21] H.-R. Song, W.-S. Choi, S.-M. Lim, and H.-D. Kim, “Target localization using RGB-D camera and LiDAR sensor fusion for relative navigation,” in *Proc. CACS Int. Autom. Control Conf. (CACS)*, Nov. 2014, pp. 144–149.
- [22] P. An et al., “Geometric calibration for LiDAR-camera system fusing 3D-2D and 3D-3D point correspondences,” *Opt. Exp.*, vol. 28, no. 2, pp. 2122–2141, 2020.
- [23] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.
- [24] J. Redmon and A. Farhadi, “YOLO9000: Better, faster, stronger,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7263–7271.
- [25] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, “YOLOv4: Optimal speed and accuracy of object detection,” 2020, *arXiv:2004.10934*.
- [26] X. Zhu, S. Lyu, X. Wang, and Q. Zhao, “TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2778–2788.
- [27] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards real-time object detection with region proposal networks,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
- [28] R. Girshick, “Fast R-CNN,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
- [29] G. Gkioxari, J. Johnson, and J. Malik, “Mesh R-CNN,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9785–9795.
- [30] C. R. Qi, H. Su, K. Mo, and L. J. Guibas, “PointNet: Deep learning on point sets for 3D classification and segmentation,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 652–660.
- [31] H. Lai, P. Yin, and S. Scherer, “AdaFusion: Visual-LiDAR fusion with adaptive weights for place recognition,” *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 12038–12045, Oct. 2022.
- [32] Y. Lu, F. Yang, F. Chen, and D. Xie, “PIC-Net: Point cloud and image collaboration network for large-scale place recognition,” 2020, *arXiv:2008.00658*.
- [33] Y. Li et al., “Deepfusion: LiDAR-camera deep fusion for multi-modal 3D object detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2022, pp. 17182–17191.



Yongkang Xu (Graduate Student Member, IEEE) received the M.Sc. degree in mechatronic engineering from Central South University, Changsha, China, in 2021. He is now pursuing the Ph.D. degree in control science and engineering with the State Key Laboratory of Intelligent Control and Decision of Complex Systems, School of Automation, Beijing Institute of Technology, Beijing, China, where he is participating in the air-ground collaborative robots.

His research interests include environmental awareness, mobile robotics, motion control, neural networks, and autonomous navigation.

Dr. Xu is a member of the State Key Laboratory of Intelligent Control and Decision of Complex Systems.



Zhihua Chen received the M.Sc. degree in mechatronics engineering from Beijing Information Science and Technology University, Beijing, China, in 2018, and the Ph.D. degree in control science and engineering from the State Key Laboratory of Intelligent Control and Decision of Complex Systems, Beijing Institute of Technology, Beijing, in 2022.

He is currently a Teacher with the School of Information Engineering, Nanchang Hangkong University, Nanchang, China. His current

research interests include robot-legged motion control, wheel-legged motion planning, and obstacle avoidance.

Dr. Chen is a member of the State Key Laboratory of Intelligent Control and Decision of Complex Systems.



Chencheng Deng received the B.S. degree from the Department of Automation, Beijing Institute of Technology, Beijing, China, in 2021, where he is now pursuing the M.S. degree in control science and engineering with the State Key Laboratory of Intelligent Control and Decision of Complex Systems, School of Automation.

His research interests include computer vision, LiDAR, neural networks, environmental awareness, and multisensor fusion.

Mr. Deng is a member of the State Key Laboratory of Intelligent Control and Decision of Complex Systems.



Shoukun Wang received the B.S., M.S., and Ph.D. degrees from the Department of Automation, Beijing Institute of Technology, Beijing, China, in 1999, 2002, and 2004, respectively.

He has joined the Department of Electronics and Computer Engineering, Purdue University, West Lafayette, IN, USA, as a Visiting Scholar. He has been teaching with the School of Automation, Beijing Institute of Technology, since 2004. His research interests include sen-

sors, measurement, and electrohydraulic control.



Junzheng Wang received the Ph.D. degree in control science and engineering from Beijing Institute of Technology, Beijing, China, in 1994.

He is the Deputy Director of the State Key Laboratory of Intelligent Control and Decision of Complex Systems, the Director of the Key Laboratory of Servo Motion System Drive and Control, and the Dean of Beijing Institute of Technology, where he is a Professor and a Ph.D. Supervisor. His current research interests include motion control, electric hydraulic servo

system, and dynamic target detection and tracking based on image technology.

Prof. Wang is the Standing Director of Chinese Association of Automation, and a Senior Member of the Chinese Mechanical Engineering Society and the Chinese Society for Measurement. He received the Second Award from the National Scientific and Technological Progress (No. 1) of China.